Dorian Koch (he/him)

Curriculum Vitae

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Education

2022–2025 **Computer Science B.Sc.**, *RWTH Aachen*, *Grade 1.0 with distinction (best possible, only achieved by 2%)*

2025–present **Computer Science M.Sc.**, *RWTH Aachen*, *Expected graduation date: March 2028*

Experience

2024–present **Student Research Assistant**, *RWTH Aachen University*

- Research in the ML/ASR i6 group (Schlüter, Ney): Speech Recognition, Denoising Language Models and Speech LLMs; first-author arXiv preprint.
- Built PyTorch/RETURNN training pipelines, ran experiments totaling 40k H100 GPU-hours.

2023–2024 **Student Tutor**, *RWTH Aachen University*

- Coordinated with co-tutors to prepare and teach all four first-semester courses for international students.

Projects

2022–2023 **MemScanner**, *C++, intrinsics (SSE, AVX, AVX2)*

- Vectorized wildcard pattern search with AVX2: 5.1x faster than scalar implementation (`std::search`).
- Saturates RAM bandwidth with 3 threads, while the scalar baseline reaches only 70% using all 8 cores.

2024–present **RWTH University gym occupancy tracker**, *TypeScript, Python, Docker*, public at `https://rwtf.dorianko.ch`

- Docker stack using Express.js, MariaDB, nginx; running for over 2 years with over 270k data points.
- Ingest scraped data via a lightweight self-built OCR stack that uses no extra dependencies.
- Implemented Nearest-neighbour forecasting: rescale past days to today's time-series via least squares; compute uncertainty bands from quantiles of the closest n days.

Volunteering

2024–2026 **VideoAG**, *RWTH Aachen University*, Student Council I/1, public at `https://rwth.video/`

- Upgraded and maintained the student council's lecture video platform in a team of two.
- Migrated from a server-rendered site to a static Next.js frontend and API backend: 3x faster time-to-interactive.
- Replaced an Ansible-based deployment with a fully containerized Docker/Kubernetes setup.

Honors, Awards & Competitions

2026 **Bachelorpreis (Bachelor's Thesis Award)**, *Faculty of Computer Science, RWTH Aachen University*
Awarded at Tag der Informatik 2026 for an outstanding Bachelor's thesis on "Denoising Language Models for Speech Recognition". Given to 4 out of 473 graduates (top 1%).

2026 **Winner, Game Jam 2026**, *Student Council I/1, RWTH Aachen University*
48-hour hackathon to code and design a game from scratch, 3-person team; 1st of 19 teams by jury vote.

2024–2026 **Competitive CTF player (crypto & rev)**, *Pwn-la-Chapelle (RWTH Aachen team)*
Participated in CSAW'24, DeadSec, hack.lu, GPN, and more; team results include 1st European team at CSAW'24 Finals.

Publications

arXiv preprint *Reproducing and Dissecting Denoising Language Models for Speech Recognition*, D. Koch, A. Zeyer, N. Rossenbach, R. Schlüter, H. Ney, 2025, arXiv:2512.13576

Skills

Programming	Python, C++, TypeScript
ML	PyTorch, RETURNN
Tools	Linux, Docker, Kubernetes, CMake, Git, CI (GitHub Actions)
Languages	German (native), English (C1), Spanish (B1)